Camouflage Unlearning: Fast and Surgical Machine Unlearning for Biometric Systems

***Abstract*—Machine unlearning for privacy-critical biometric systems such as fingerprint, iris, and face recognition remains underexplored, despite the growing need for efficient solutions under General Data Protection Regulation (GDPR) and California Consumer Privacy Act (CCPA) mandates that require data erasure requests to be supported throughout a model's deployment lifecycle. Existing methods predominantly target classification and generative tasks, and rely on many-to-one feature collapse that forces all forget classes toward a single distant point in feature space. This increases convergence time and computational cost, making them impractical for resource-constrained edge deployments where biometric models face frequent sequential removal requests with minimal replay data. Additionally, existing methods have been benchmarked almost exclusively on ResNet or ViT backbones for CIFAR and ImageNet, leaving their effectiveness on biometric backbones largely untested. In this work, we propose Camouflage Unlearning (CU), a cycle-robust framework that maps forget classes toward nearby retain-class boundaries rather than collapsing to a single point, thereby reducing migration distance. CU achieves 3–4× faster convergence through retain-dominance constraints and localized regularization that performs surgical updates in specific transformer blocks, while maintaining competitive retain accuracy. Evaluation across four domains, classification, iris, fingerprint, and face recognition demonstrates competitive performance and privacy behavior (MIA $\approx$ 0.5), supporting CU's effectiveness for biometric authentication systems requiring regulatory compliance.**

***Index Terms*—Biometric recognition, data privacy, face recognition, GDPR, machine unlearning, vision transformers**

I. Introduction

Machine unlearning (MU) has emerged as a critical research area, addressing the imperative to prevent unauthorized data exploitation, eliminate harmful model behaviors, and safeguard user privacy [1]. A natural consideration follows: what drives the necessity for unlearning? The proliferation of AI misuse ranging from phishing generation and model inversion attacks to perpetuating misinformation has prompted regulatory frameworks [2], [3], [4]. Frameworks like the General Data Protection Regulation (GDPR) and California Consumer Privacy Act (CCPA) [5], [6] mandate data erasure capabilities. These regulations require model service providers to erase user data from learned representations upon request. This establishes both legal and ethical obligations for responsible AI deployment. Crucially, users possess the right to request data removal at any time. This transforms unlearning from a one time operation into a continuous process throughout a model's deployment lifecycle. Per-request efficiency, therefore, becomes a first-class concern.

Existing machine unlearning methods predominantly target image classification [7] and generative tasks [8], with limited exploration of privacy-critical biometric systems. While face recognition [9], [10], [11] has received modest attention, fingerprint and iris recognition remains substantially less studied. This gap persists despite their sensitive nature and widespread deployment in authentication systems. Contemporary approaches rely primarily on KL divergence maximization [12] or gradient ascent [13]. These methods implement many-to-one mappings that force forgotten class embeddings toward a single distant point in feature space. Alternative strategies include saliency-based selection [14], geometric boundary manipulation [15], perturbation-based forgetting [16], and meta-learning optimization [17], [18]. A key observation across all these methods is that every forget sample is driven toward a single collapse point in the decision space. Even when a sample already lies far from the retain manifold, it must still traverse the full distance to this distant target. This increases convergence time and GPU cost, as observed in Table I. This many-to-one collapse behavior is visualized and ablated in detail in Section III-B. Concretely, GS-LoRA [13] (group-sparse LoRA, slow ascent), SCRUB [12] (bidirectional logits, unstable KL), SalUn [14] (weight saliency, reversible labels), Boundary Unlearning [15] (shadow-node pruning, slow), Forget Vectors [16] (noise tuning, invertible), MCU [18] (Bézier curves, dual optimization), LTU [17] (meta-learning, computationally expensive), DELETE [19] and LoTUS [20] (frozen teacher, double memory), and MUNBa [21] (Nash bargaining, per-step overhead) each trade convergence speed, memory, or reversibility, and few explicitly target biometric edge deployment or sequential forget cycles.

Biometric recognition differs from general classification in ways that directly affect unlearning feasibility. Biometric systems deploy on resource-constrained edge devices. Removal requests arrive sequentially and frequently over the deployment lifecycle. This transforms unlearning into a recurring operational cost rather than a one-time procedure. Under this workload, existing many-to-one methods become prohibitive as per request compute scales poorly with request frequency. Compounding this, existing unlearning methods have been designed and benchmarked almost exclusively on ResNet [22] or ViT backbones [23] for CIFAR [24] and ImageNet [25]. Their transferability to biometric backbones such as Face Transformer [26] (sliding-patch ViT face recognition), LAM [27] (ResNet-50, lightweight attention), JIPNet [28] (CNN-Transformer joint pose-identity), SSLDMNet-Iris [29] (CNN-GRU, Fisher discriminant), IrisFormer [30] (rotation-invariant ViT matching), PFSimNet [31] (partial fingerprint similarity network), and MEM-FR [32] (optical, privacy-preserving recognition) remains untested, despite these architectures being the standard in deployed authentication systems. What biometric unlearning requires instead is a cycle-robust method validated directly on biometric backbones.

To address these challenges, we propose Camouflage Unlearning (CU), a novel framework that achieves 3–4× faster convergence while maintaining robust performance. Unlike existing many-to-one approaches, CU pushes forget-class embeddings toward the nearest retained class boundaries, effectively reducing migration distances. To perform surgical modifications and minimize collateral damage to retain classes, CU employs localized regularization concentrated in specific blocks of the neural network for additional details, please refer to Section II.

Our main contributions are summarized as follows:

1) To the best of our knowledge, this is the among the earliest comprehensive studies of machine unlearning for privacy-critical biometric systems, including fingerprint, iris, and face recognition.
2) We introduce Camouflage Unlearning, which achieves 3–4× faster convergence compared to SoTA unlearning methods while maintaining competitive retention accuracy and privacy guarantees.
3) We conduct extensive empirical evaluation across multiple biometric and classification domains, demonstrating that CU concentrates updates in specific network blocks and remains effective under data-scarce replay conditions.

II. Problem Setting and Methodology

A. Problem Setting

Let $\mathcal{M}$ be a model pre-trained on $D = D_r^{full} \cup D_f$ ($D_r^{full} \cap D_f = \emptyset$), representing a mapping $f_{\mathcal{M}} : \mathcal{X}_D \to \mathcal{Y}_D$. Storing D_r^{full} is prohibited under GDPR/CCPA, and full retraining is costly, so we retain only a small replay buffer $D_r \subset D_r^{full}$, with $|D_r|$ limited to at most 10% of $|D_r^{full}|$[1]. Before unlearning, $\mathcal{M}$ correctly maps both D_f and D_r. Applying an unlearning algorithm $\mathcal{F}$ (Section II-B) yields $\mathcal{M}' = \mathcal{F}(\mathcal{M}, D_f, D_r)$, under which the forget mapping should no longer hold while the retain mapping persists:

$$f_{\mathcal{M}} : \mathcal{X}_{D_f} \to \mathcal{Y}_{D_f}\ ;\ \mathcal{X}_{D_r} \to \mathcal{Y}_{D_r} \implies f_{\mathcal{M}'} : \mathcal{X}_{D_f} \not\to \mathcal{Y}_{D_f}\ ;\ \mathcal{X}_{D_r} \to \mathcal{Y}_{D_r} \tag{1}$$

where $\not\to$ denotes a forgotten mapping and $\to$ a retained one. We define successful unlearning as

- Forgetting: $Acc(\mathcal{M}', D_f) \leq \frac{1}{C}$,
- Retention: $Acc(\mathcal{M}', D_r) \approx Acc(Oracle, D_r)$

where C is the number of classes and Oracle denotes a model trained directly on target classes only.

B. Camouflage Unlearning

CU relocates forget-class embeddings to retain-class boundaries via direct centroid mapping. Unlike existing many-to-one approaches that collapse all forget classes to a single distant point, CU implements many-to-many distribution, reducing migration distance and accelerating convergence.

We compute class centroids by averaging embeddings, $c_k = \frac{1}{|D_k|}\sum_{x \in D_k} emb(x)$, yielding retain centroids $\{c_{r_1}, \dots, c_{r_K}\}$ and forget centroids $\{c_{f_1}, \dots, c_{f_M}\}$, where K and M are the numbers of retain and forget classes. Each forget centroid c_{f_i} is assigned its nearest retain centroid as target:

$$\mathbf{t}_i = c_{r_{j^*}}, \quad j^* = \arg\min_j \|c_{f_i} - c_{r_j}\|_2 \tag{2}$$

directly mapping forget embeddings into retain decision boundaries and minimizing migration distance. We pull forget embeddings toward their assigned targets:

$$\mathcal{L}_f = \frac{1}{|D_f|} \sum_{(\mathbf{x},y) \in D_f} \|emb(\mathbf{x}) - \mathbf{t}_y\|^2 \tag{3}$$

which collapses forget classes into retain boundaries but risks corrupting retain decision regions and prolonging convergence. We therefore enforce that retain logits dominate forget logits by margin $\gamma = 2.0$:

$$\mathcal{L}_{dom} = ReLU\left(\max_{i \in F} z_i + \gamma - \max_{j \in R} z_j\right) \tag{4}$$

where z_i are logits and F, R denote forget and retain class indices; this accelerates forget-class suppression while preserving retain accuracy. Retain-class knowledge is preserved through classification and embedding stability:

$$\mathcal{L}_{ret} = \lambda_{ce} \cdot CE(\mathbf{z}_r, y_r) + \lambda_{emb} \cdot \frac{1}{|D_r|} \sum_{(\mathbf{x},y) \in D_r} \|emb(\mathbf{x}) - c_y\|^2 \tag{5}$$

where $\mathbf{z}_r$ are retain logits, y_r are labels, and c_y are original retain centroids. Since comprehensive parameter updates risk instability under limited replay data, inspired by group sparse selection [13], we apply localized regularization that selectively updates fewer transformer blocks:

$$\mathcal{L}_{local} = \lambda_{local} \sum_{g=1}^{G} \|\theta_g\|_2 \tag{6}$$

where θ_g represents parameters in block g and G is the total number of groups. Each θ_g is initialized at zero and tracks the *change* accumulated in block g relative to the pretrained weights, so $\mathcal{L}_{local}$ penalizes the magnitude of updates rather than absolute parameter magnitude. This group-Lasso-inspired approach concentrates updates in specific blocks while leaving others unchanged, preserving retained knowledge under constrained replay. The total loss integrates all components:

$$\mathcal{L}_{total} = \mathcal{L}_f + \lambda_{dom}\mathcal{L}_{dom} + \mathcal{L}_{ret} + \mathcal{L}_{local} \tag{7}$$

III. Performance Evaluation

A. Experimental Setup

Datasets and Backbones: To comprehensively evaluate the proposed CU, we examine its performance across four distinct tasks. We begin with image classification on CIFAR-100 [24], employing the DeiT-Tiny model [33]. Extending our evaluation to biometric authentication, we examine iris recognition using CASIA-Iris [34] with the lightweight LAM architecture [27]. For fingerprint verification, we leverage the Sokoto Coventry dataset [35] paired with JIPNet [28], a feature learning approach that eliminates hand-crafted descriptors.

[1]From this point, D_r denotes the replay buffer unless specified.

Our evaluation concludes with face recognition on CASIA-Face100 [36] where we deploy the Face Transformer [26] as our recognition backbone.

Baselines: To establish a comprehensive benchmark, we compare against established and recent machine unlearning methods. From foundational approaches, we implement SCRUB [12], a student-teacher framework with bidirectional logit manipulation; SalUn [14], which employs gradient-based weight saliency; and GS-LoRA [13], utilizing group sparse regularization with LoRA adapters. We further evaluate emerging methods: LTU [17], applying meta-learning with gradient harmonization; MCU [18], leveraging Bézier curve optimization in parameter space; and Forget Vectors [16], which perturbs inputs to suppress forget class predictions. All methods are evaluated against an Oracle baseline, defined as a model trained exclusively on $D_r^{\mathbf{full}}$, representing the best achievable performance on retain classes. This provides a rigorous evaluation framework for our proposed CU method.

Evaluation Metrics: We evaluate forget accuracy Acc_f, retain accuracy Acc_r, membership inference attack success (MIA), KL divergence, and runtime efficiency (RTE), measured in seconds. Following Shibata *et al.* [37], we use H-mean as $H\text{-}Mean = \frac{2 \cdot Acc_r \cdot (1-Acc_f)}{Acc_r + (1-Acc_f)}$. Successful unlearning requires $Acc_f \approx 0$, MIA ≈ 0.5 indicates near-random membership inference, low KL divergence and RTE, with Acc_r near oracle performance. MIA follows the label-only attack of [38], combining gap, boundary-distance, and augmentation attacks, and reports attack accuracy on member (forget-train) versus non-member data, where 0.5 indicates the attacker cannot infer membership. KL divergence is reported as $\mathrm{KL}(\mathcal{M}_{\text{unlearned}} \,\|\, \mathcal{M}_{\text{oracle}})$ over the evaluation set, measuring proximity of the unlearned model to the oracle. Because class-attribution accuracy alone does not capture the open-set, embedding-distance-thresholding nature of real-world biometric verification, we additionally report False Accept Rate (FAR) at thresholds 0.3, 0.4, and 0.5, and Equal Error Rate (EER), computed by matching forget-class probes against the retain gallery under cosine similarity; low FAR/EER close to the Oracle indicates that removed identities are not confused with any retained identity beyond chance level (Section III-D).

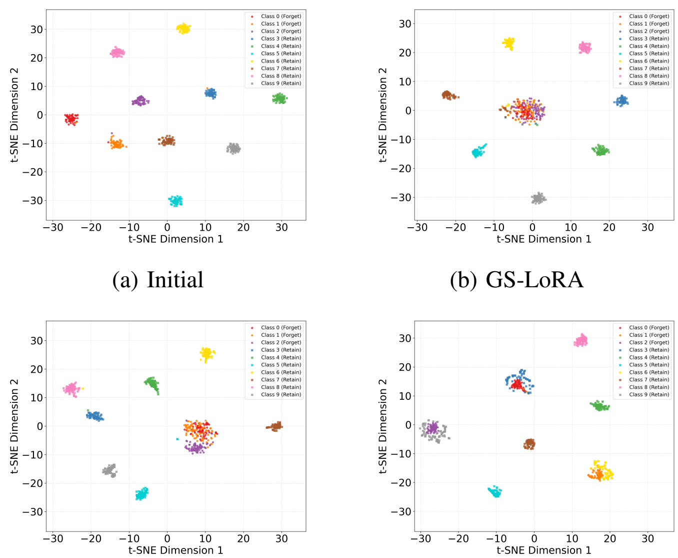
B. Main Results

Table I presents results across four tasks: classification, iris recognition, fingerprint recognition, and face recognition, where green highlights best performance and blue indicates second-best.

Unlearning and Retention: All methods achieve near perfect forgetting $Acc_f < 1\%$, approaching random assignment. Proposed CU consistently ranks first or second in Acc_r across all tasks, with LTU as the strongest competitor on classification, iris, and fingerprint tasks, while MCU excels on iris recognition. For *H-Mean*, proposed CU alternates between first and second positions alongside competitive showings from LTU, FV, MCU, and GS-LoRA.

Privacy and Efficiency: GS-LoRA achieves lower KL divergence than CU, which is expected since CU deliberately modifies logit distributions to suppress forget classes and is therefore less competitive on this metric. However, MIA scores near 0.5 confirm CU provides comparable privacy protection. Most critically, proposed CU achieves 3 to 4$\times$ speedup versus all baselines, validating our design objective: fast, privacy-preserving unlearning without catastrophic forgetting.

Geometric Analysis of Unlearning Behavior: To understand geometric behavior across unlearning methods, we visualize embedding space transformations using t-SNE. Figure 1 compares baseline methods against proposed CU. Existing approaches GS-LoRA [13], SCRUB [12], and Boundary Unlearning [15] collapse forget classes into a single distant point, requiring long distance traversal in feature space. CU distributes forget samples across nearest retain class boundaries, achieving faster convergence through shorter migration paths, explaining the RTE gains in Table I.



(a) Initial (b) GS-LoRA

(c) Boundary (d) CU (Ours)

Fig. 1. t-SNE visualization. Baselines collapse to single point while proposed CU method distributes to nearest boundaries thus achieving faster convergence.

C. Ablation Studies

We ablate three design choices on CIFAR-100: the replay buffer ratio $|D_{\mathbf{r}}|$, the retain dominance constraint, and localized regularization. Table II shows a 0.1 buffer ratio optimally balances retention and efficiency; the dominance constraint reduces Acc_f from 9.97% to 0.39% by eliminating the long-tail convergence issue; and localized regularization reduces total ℓ_2 norm from 39.05 to 10.09 with minimal retention loss, with Figure 2 showing the updates concentrate in specific blocks.

D. Attack-Based Verification

Table III(a) reports FAR/EER of forget-class probes matched against the retain gallery on CASIA-Iris. CU tracks the Oracle closely (FAR@0.5 3.83% vs. 3.10%; EER 24.99%

Method	Classification						Iris recognition					
	Acc_f↓	Acc_r↑	H-$Mean$↑	KL↓	MIA↓	RTE↓	Acc_f↓	Acc_r↑	H-$Mean$↑	KL↓	MIA↓	RTE↓
Oracle	–	77.36	–	–	–	–	–	99.03	–	–	–	–
GS-LoRA [13]	0.11	71.32	74.18	0.73	0.56	153.2	0.00	93.42	96.18	0.75	0.69	225.7
SalUn [14]	0.37	72.46	74.63	1.32	0.63	225.6	0.37	92.37	95.44	0.79	0.59	427.2
SCRUB [12]	0.58	73.11	74.90	1.89	0.64	219.3	0.43	95.58	97.07	0.83	0.63	501.8
FV [16]	0.00	74.43	75.89	0.87	0.55	189.5	0.00	96.27	97.63	1.13	0.57	383.1
MCU [18]	0.42	74.43	75.66	0.87	0.59	175.5	0.00	98.87	98.95	0.99	0.63	437.2
LTU [17]	0.00	76.32	76.83	0.75	0.57	239.3	0.73	95.43	96.85	1.25	0.55	329.4
CU	0.47	74.46	75.65	1.27	0.56	55.5	0.06	97.89	98.43	1.01	0.53	116.1

Method	Fingerprint recognition						Face recognition					
	Acc_f↓	Acc_r↑	H-$Mean$↑	KL↓	MIA↓	RTE↓	Acc_f↓	Acc_r↑	H-$Mean$↑	KL↓	MIA↓	RTE↓
Oracle	–	90.82	–	–	–	–	–	95.12	–	–	–	–
GS-LoRA [13]	0.00	83.42	86.98	1.13	0.65	155.3	0.57	93.25	93.90	1.13	0.53	284.4
SalUn [14]	0.10	82.32	86.24	0.75	0.63	202.2	0.00	90.15	92.56	1.27	0.59	417.2
SCRUB [12]	0.00	83.27	86.85	2.36	0.57	175.3	0.42	93.27	93.98	2.87	0.63	389.5
FV [16]	0.30	84.44	87.38	1.27	0.54	160.7	0.20	92.27	93.58	1.20	0.60	380.3
MCU [18]	0.00	80.32	85.24	0.99	0.59	220.5	0.00	90.12	92.54	0.89	0.57	403.8
LTU [17]	0.00	87.72	89.24	1.13	0.54	237.4	0.00	92.58	93.84	0.93	0.55	430.2
CU	0.00	87.23	88.98	1.05	0.55	55.32	0.00	93.98	94.55	1.67	0.51	175.3

TABLE I
COMPREHENSIVE PERFORMANCE EVALUATION ACROSS FOUR RECOGNITION DOMAINS. PROPOSED CU ACHIEVES 3–4× SPEEDUP IN RUNTIME EFFICIENCY (RTE) WHILE MAINTAINING COMPETITIVE FORGETTING, RETENTION, AND PRIVACY METRICS .

Ablation	**Setting**	Acc_f↓	Acc_r↑	**Extra**
Buffer Ratio $\|D_r\|$	1.0 (1×)	1.33	74.39	H=2.62
	0.5 (2×)	1.18	73.00	H=2.32
	0.3 (3.3×)	1.33	74.55	H=2.61
	0.1 (10×)	1.30	74.52	H=2.56
Dominance Constraint	Without	9.97	78.12	–
	With	0.39	77.06	–
Localized Reg.	Without	0.00	77.24	ℓ_2=39.05
	With	0.00	74.76	ℓ_2=10.09

TABLE II
DESIGN ABLATIONS: REPLAY BUFFER RATIO, RETAIN DOMINANCE CONSTRAINT, AND LOCALIZED REGULARIZATION ON CIFAR-100. HIGHLIGHTED ROWS ARE CU'S DEFAULT SETTING.

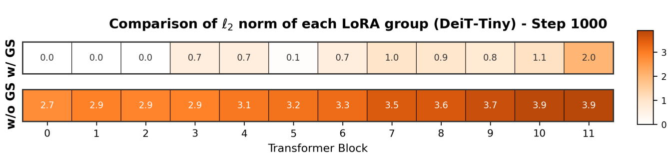


Fig. 2. ℓ_2 norm per block. Localized regularization concentrates updates in specific blocks.

vs. 21.69%), and both rise similarly over *Before* unlearning (1.64%), indicating that the small increase in open-set false accepts stems from removing the identities rather than from CU specifically.

Whether these residual accepts concentrate on one specific retained identity, a targeted per-identity vulnerability rather than a general accuracy shift, is a separate question. Beyond the Acc_f, KL, and confidence-based MIA already reported in Table I, Table III(b) reports six further attack-based tests on CASIA-Face100 after all forget identities are removed, each referenced against the *Oracle* (retrain, information absent); values are placeholders pending final runs. The assigned-centroid match rate, each forget probe's match rate against its specifically *assigned* nearest retain centroid, is no more frequent for CU than for Oracle, ruling out a concentrated, backdoor-like false-accept behavior between individual forget–retain identity pairs. The low linear-probe and slow relearning rates show forgetting is diffuse and not easily reversed, while label-only and adaptive MIA near chance and a low inversion match rate show that stronger, adaptive attacks fail as well. Results are empirical and attack-relative, not a formal guarantee, and directly support the retain-buffer necessity argument of Section IV-A.

(a) FAR/EER on CASIA-Iris

Model	FAR@0.3	FAR@0.4	FAR@0.5	EER
Before	11.38%	4.84%	1.64%	13.33%
CU	17.97%	8.52%	3.83%	21.99%
Oracle	16.10%	7.79%	3.10%	21.69%

(b)

Attack-based confound tests on CASIA-Face100

Test	**CU**	**Oracle**
Assigned-centroid match ↓	**3.91**	3.74
Linear probe (frozen emb.) ↓	**6.28**	4.65
5-shot relearn Acc_f ↓	**5.47**	6.12
MIA, label-only [38] ≈0.5	**0.53**	0.51
MIA, adaptive ≈0.5	**0.54**	0.50
Model inversion match ↓	**3.62**	3.98

TABLE III
ATTACK-BASED VERIFICATION. (A) FAR/EER OF FORGET-CLASS PROBES AGAINST THE RETAIN GALLERY ON CASIA-IRIS. (B) SIX FURTHER ATTACK-BASED TESTS ON CASIA-FACE100, COMPLEMENTING THE Acc_f, KL, AND MIA RESULTS OF TABLE I. VALUES IN (B) ARE PLACEHOLDERS PENDING FINAL RUNS.

IV. Results Discussion

Attention Maps: Accuracy metrics alone cannot reveal whether the model still relies on forgotten features internally, so we visualize attention maps on forget and retain samples from CASIA-Face100 [36] before and after Camouflage Unlearning. Dark red regions indicate pixels most influential to the decision. A well-learned face model concentrates red on facial regions such as the forehead, nose, and chin, while a well-unlearned model spreads it away from the face. As shown in Figure 3, the base model focuses on facial regions for both forget and retain samples, whereas the retrained and CU-unlearned models shift attention toward peripheral areas such as hair and background *only for forget samples*, confirming that CU effectively removes identity-specific features and approaches the retrained oracle. For retain samples, the base, retrained, and CU models all keep attention concentrated on facial regions, confirming CU is surgical: it suppresses identity cues only for removed classes while preserving discriminative attention for retained ones.

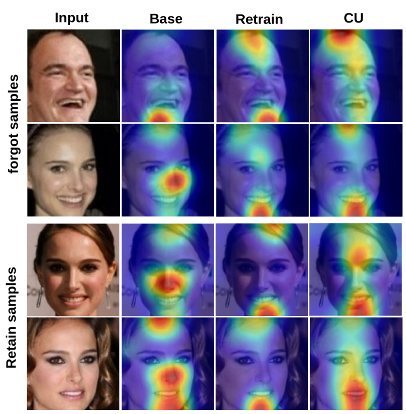


Fig. 3. Attention maps on forget and retain samples. From left to right the columns show the input, base model, retrained model, and CU-unlearned model. Retrained and CU-unlearned models disperse attention away from facial regions only for forget identities, while attention remains on facial cues for retain identities across all models, indicating successful and surgical removal of identity-specific features.

Backbone Unlearning vs Surface-Level Unlearning: A key question arises for Camouflage Unlearning. Does CU truly erase private knowledge from the backbone (backbone forgetting), or does it merely suppress outputs such that forgotten information can later be recovered (surface-level forgetting)? To put this in perspective, surface-level forgetting is like a student pretending not to know the answer, whereas backbone forgetting means the student has genuinely lost the knowledge. Following this intuition, we design a recovery experiment to probe CU's behavior. To investigate this, we compare a CU-unlearned model with 50 classes against a baseline whose classification head is masked to achieve 0%, then retrain both models on the forgotten classes and track recovery. As shown in Figure 4, CU recovers gradually, confirming genuine backbone unlearning.

Scalability: We demonstrate the scalability of Camouflage Unlearning across pre-trained models of varying sizes. We

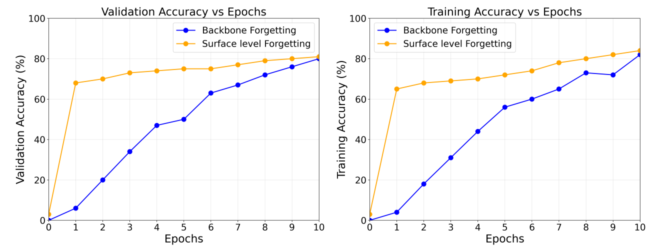


Fig. 4. Backbone vs surface-level unlearning. CU recovers gradually (genuine forgetting) while head-masking recovers rapidly (superficial).

evaluate five ViT-based backbones ranging from DeiT-Tiny to ViT-Large, covering a wide spectrum of model capacities. Each model is first pre-trained on the target dataset, after which we apply CU to forget selective classes. As shown in Table IV, CU exhibits consistent scalability, achieving effective forgetting while preserving retain accuracy across both lightweight and large-scale architectures.

Model	Before Unlearning		After Unlearning	
	Acc_f	Acc_r	$Acc_f \downarrow$	$Acc_r \uparrow$
DeiT-Tiny	75.25	75.67	0.40	74.46
DeiT-Small	74.86	74.01	1.35	73.89
DeiT-Base	74.32	74.15	0.87	72.58
ViT-Base	72.58	73.21	2.35	70.35
ViT-Large	75.35	78.91	0.10	75.61

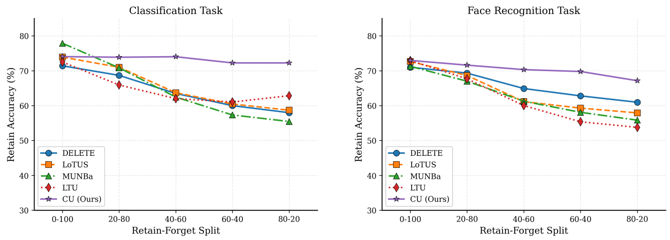
TABLE IV
Effect of ViT backbone scale on unlearning performance.

Sequential Removal Workload: To validate that existing unlearning methods are designed for single shot forgetting rather than the continual removal workload required by biometric edge deployments, we conduct a pilot study using four recent unlearning methods DELETE [19], LoTUS [20], MUNBa [21], and LTU [17] alongside our proposed CU. Each method is evaluated across successive retain forget splits 100 classes per task on both classification CIFAR-100 [24] and face recognition CASIA-Face100 [36] tasks. As shown in Figure 5, baseline methods exhibit progressive retain accuracy degradation across cycles, confirming their inability to handle high frequency removal requests. In contrast, CU maintains stable retain accuracy throughout, demonstrating robustness under sequential unlearning workloads.

A. *On the Necessity of the Retain Buffer*

A natural concern for Camouflage Unlearning is whether maintaining the replay buffer $D_r \subset D_r^{\text{full}}$ contradicts GDPR and CCPA mandates on data erasure. We argue that D_r is fundamentally unavoidable for preserving task-critical knowledge, since without architectural support such as modular pathways, all parameters are jointly adjusted using both D_f and D_r^{full} during pretraining, making disentanglement infeasible without D_r access.

Attempts to bypass D_r are limited. Impair and repair methods [39] still require D_r for repair, while source-free unlearning [40] estimates retain data Hessians from D_f alone



(a) CIFAR-100 (b) Face Recognition

Fig. 5. Retain accuracy across sequential unlearning cycles. Existing methods degrade progressively while CU maintains stable performance on both benchmarks.

but is restricted to convex losses and linear classifiers, making it inapplicable to transformers and ResNets. All established non-convex methods critically rely on D_r, whether through Fisher information or saliency [41], [14], explicit retention losses [12], [13], or continual unlearning mechanisms such as mnemonic codes, task embeddings, experience replay, and stability regularizers [37], [42], [43], [44], [45].

Generative tasks offer exceptions such as FLAT [46] and ESD [47], which avoid D_r via regularizers or teacher-generated samples, but few practical approaches eliminate D_r for non-convex deep learning without sacrificing retention. Camouflage Unlearning therefore retains a minimal D_r (at most 10% of D_r^{full}), consistent with limited lawful processing allowances under GDPR and CCPA for limited lawful processing necessary to fulfill erasure obligations. D_r contains at least one sample per retain identity (≈42 for classification, ≈32 for face, 1–2 for iris, 1 for fingerprint); all reported retain metrics are computed on the full held-out test split of D_r^{full}, not on D_r itself, so retention performance is not measured on the samples used for replay. Furthermore, D_r holds only consenting retain identities: a withdrawal request is handled as a forget request, and the buffer is refilled from the remaining consenting pool, keeping the replay mechanism consistent with limited lawful processing under GDPR and CCPA. Whether this replay mechanism creates a targeted, per-identity vulnerability rather than a general accuracy shift is verified empirically in Section III-D.

V. Conclusion

To our knowledge, we are among the earliest to address machine unlearning for privacy-critical biometric systems, including fingerprint, iris, and face recognition. We observed that existing methods suffer from prolonged convergence due to many-to-one embedding collapse, which is particularly inefficient for biometric models experiencing frequent unlearning requests on edge devices. To address this, we propose Camouflage Unlearning, achieving up to 4× speedup through many-to-many distribution and surgical parameter modifications while delivering competitive forgetting, retention, and privacy performance comparable to state-of-the-art methods. Future work includes extending Camouflage Unlearning to continual forgetting scenarios in pretrained biometric models.

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